Trainings

General Information

A hydraulic pump can be mounted on the front or at the rear of the gear cover.

The SAE A two-bolt type flange is required to mount a hydraulic pump at the rear of the gear cover. The SAE B four-bolt type flange is required for a hydraulic pump to be mounted on the front of the gear cover. There is a specific drive required to match the different sizes of spline drives required for the optional hydraulic pumps.

Engines that do **not** have a hydraulic pump drive **must** have parts that block the oil holes in the gear cover and housing for correct engine oil pressure. A four-bolt flange that has a blind plug welded to it **must** be installed on the front of the gear cover. The blind plug will extend into the bushing, blocking the oil drilling. A three-bolt cover that has two o-rings **must** be installed on the rear of the gear housing.

A hydraulic pump drive can be installed by removing the blind plug flange and the cover. If **only** the rear drive port is used, a plain four-bolt cover and gasket **must** be installed on the front of the cover. If **only** the front drive port is used, a two-bolt cover and gasket **must** be installed on the rear of the pump drive.

Preparatory Steps

Note : Some engines contain a cover plate instead of a hydraulic pump.

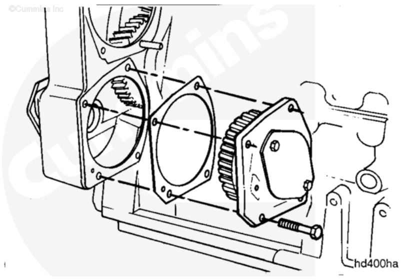
- Remove the hydraulic pump. Use the equipment manufacturer's instructions.

Remove

Remove the four hydraulic pump drive mounting capscrews.

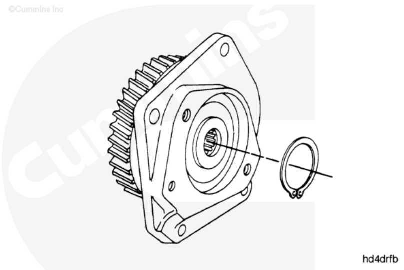
Remove the hydraulic pump drive.

Remove and discard the gasket.



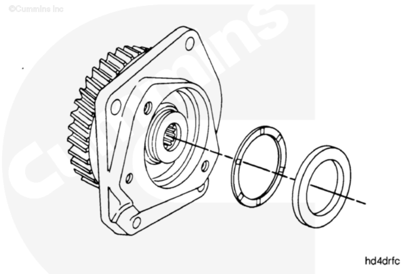
Disassemble

Remove the retaining ring.



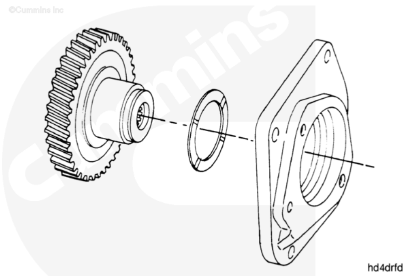
Remove the clamping washer.

Remove the thrust bearing.



Remove the shaft and gear assembly from the housing.

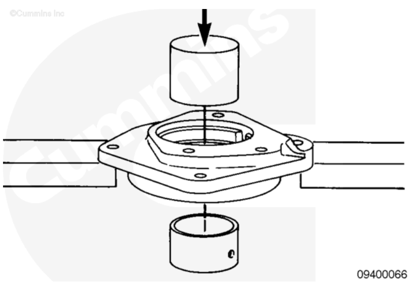
Remove the thrust bearing.



Note : Only replace the bushing if it is damaged, or is not within specifications.

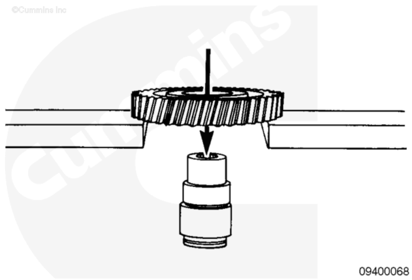
Support the housing in an arbor press.

Press the bushing out of the housing with a mandrel and arbor press.



Note : Only remove the gear from the shaft when the gear or the shaft **must** be replaced.

Remove the gear from the shaft with an arbor press.



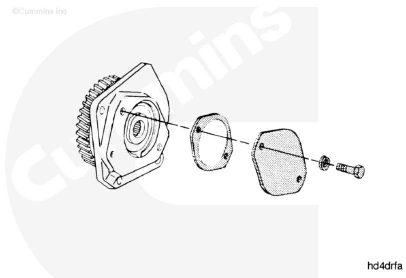
Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Clean the hydraulic pump drive housing and gear with solvent.

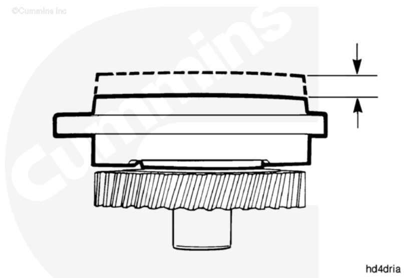
Dry with compressed air.

Remove the cover from the pump.

Remove and discard the gasket.

Measure the hydraulic pump drive end clearance.

mm		in
0.13	MIN	0.005
0.48	MAX	0.019

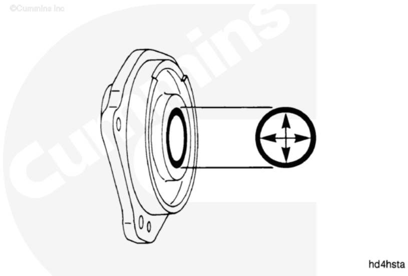


If the hydraulic pump drive is **not** within specifications, it **must** be reconditioned.

Clean the housing with solvent.

Measure the bushing inside diameter.

mm		in
50.813	MIN	2.0005
50.902	MAX	2.0050



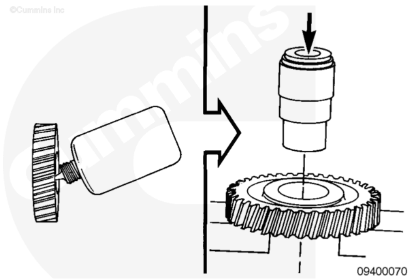
If the bushing is **not** within specifications, it **must** be replaced.

Assemble

Apply a smooth coating of Loctite® 609, or equivalent, to the inside diameter of the gear.

Support the gear in an arbor press with the part number down.

Press the shaft through the gear until the shoulder of the shaft touches the gear.

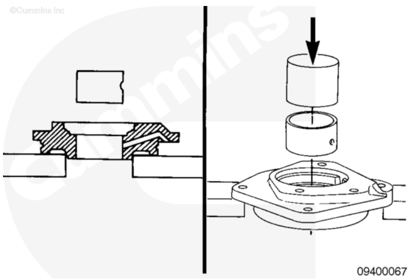


Place the housing in an arbor press.

Align the oil hole in the bushing with the oil drilling in the shaft.

Press the bushing into the housing with a mandrel and arbor press.

Both ends of the bushing **must** be positioned even with or below the surface of the housing.

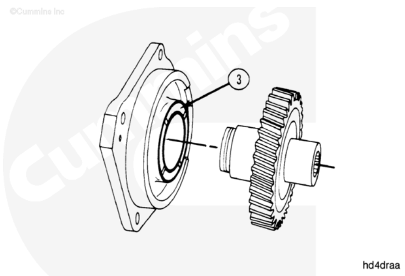


Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105, or equivalent.

Position the grooved surface of the thrust bearing (3) as shown in the illustration.

Slide the gear and shaft assembly into the housing.

The gear **must** touch the thrust bearing.

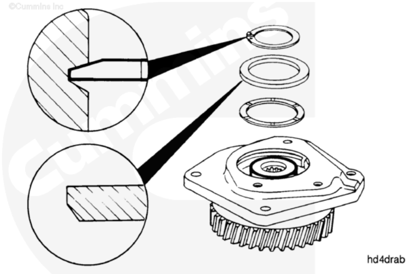


Lubricate the grooved surface of the thrust bearing with Lubriplate™ 105 or equivalent.

Slide the thrust bearing over the shaft with the grooved side positioned up.

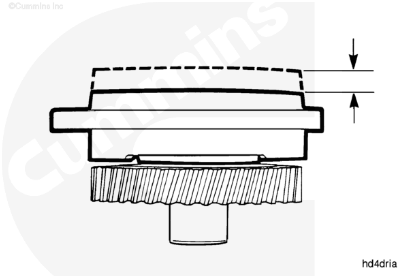
Slide the clamping washer over the shaft with the beveled edge positioned against the thrust bearing.

Install the retaining ring with the beveled edge positioned as shown in the illustration.



Measure the hydraulic pump drive end clearance.

mm		in
0.13	MIN	0.005
0.48	MAX	0.019

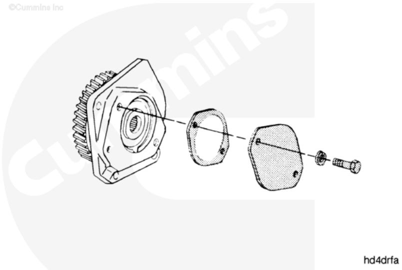


If the hydraulic pump drive is **not** within specifications, it **must** be reconditioned.

Install a new gasket.

Install the cover.

Install the lock washers and capscrews.



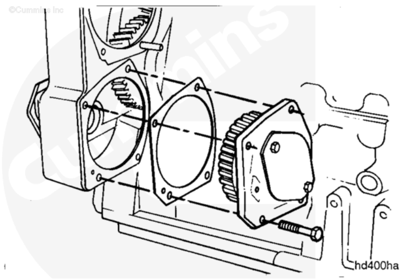
Install

The hydraulic pump drive or cover plate for engines with a two-piece front cover **must** have an o-ring, in addition to the gasket.

Lubricate the bushing in the front cover with clean engine oil.

Install the gasket, hydraulic pump drive, lock washers, and capscrews.

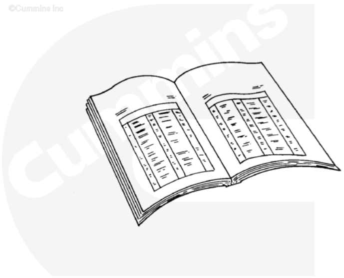
Tighten the capscrews.



Torque Value: 45 n•m [33 ft-lb]

Finishing Steps

- Install the hydraulic pump. Refer to the OEM troubleshooting and repair manual.
- Operate the engine and check for leaks.



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Last Modified: 18-Nov-2008